

Deinking of Water-Based Ink Printing from Plastic Film Using Nonionic Surfactants

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ABSTRACT: When plastic film packaging is processed for reuse, residual ink colors the polymer and makes the polymer less stiff, weaker, and denser than the original material. These problems can be avoided if the printing is removed from the plastic film surface. In this study, a commercial polyethylene film with water-based ink printing was deinked (the ink was removed) using nonionic surfactants under a variety of conditions, and the effect of some parameters related to the deinking process were quantified and optimized. Nonylphenol polyethoxylate, ethoxylated alcohol at a pH of 10, and ethoxylated amine at either acidic, neutral, or basic pH level were identified to be effective at deinking. The hydrophilic–lipophilic balance (HLB) value and the cloud point are useful parameters for characterizing the effectiveness of nonionic surfactants. For optimal deinking, the HLB of the nonionic surfactants should be as small as possible or the cloud point as low as possible without the deinking temperature substantially exceeding the cloud point. Information related to scaleup of the process was also developed. Soaking plastic film in surfactant solution prior to mechanical agitation significantly decreased the needed agitation time to achieve a given level of deinking. Deinking decreases as the plastic/solution ratio (consistency) increases, and cutting plastic film into smaller pieces improves the deinking at moderate consistency. Increasing agitation speed also enhances the deinking. Experiments performed with pilot-sized paper deinking apparatuses demonstrated that deinking of plastic film using surfactants is technically feasible.

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KEY WORDS: Color measurement, deinking of plastic film, HLB, nonionic surfactant, plastic film recycling, tensile properties, water-based ink.

Environmental and solid waste disposal considerations along with shortages of raw material for plastic resins have increased the need for plastic film recycling. Currently,

approximately 9% by weight and 20% by volume of U.S. municipal solid waste is composed of plastics; of this, 21 to 57% is plastic film (1–3). Only about 3.5% of all plastics and 5% of plastic films are now recycled (1,4).

Replacement of plastic in packaging by other materials such as paper, foil, and glass has been considered as an alternative to reduce plastic waste, but this would result in a 400% increase in the weight of waste, a 200% increase in energy consumption in making the alternatives, and a 210% increase in cost (2). Plastic packaging improves product security and reduces spoilage and contamination of food products. The market demand for plastic film packaging in the United States is projected to grow at an annual rate of 3.1%, which will also increase the amount of plastic film requiring disposal or recycle (5). Limitations in the technology of sorting, cleaning, and reformulating the recyclate limit the plastic recycling rate. Since the source is better defined, in-plant plastic recycling is more successful than recycle of post-consumer plastic. However, if the plastic film is printed, re-extrusion of this plastic results in a product that can only be used to make low-value colored products such as trash cans, pipes, and signposts. Repelletized colored resins can crack owing to gases associated with ink decomposition (3,6,7). Effective film deinking technology could permit reuse of the polymer to produce clear films. Possible alternatives for plastic film deinking include using either organic solvents or surfactants. However, surfactants have an advantage over organic solvents since the former are more environmentally compatible (6).

Polymers used for films include low-density polyethylene (LDPE), high-density polyethylene (HDPE), polyethylene terephthalate, ethylene vinyl acetate, polyvinyl chloride, and polypropylene. The highest-volume resin used is LDPE because of its toughness, sealability, inertness, and action as a moisture barrier. LDPE has greater clarity and flexibility than HDPE, making it particularly valuable for clear films. Most plastic packaging is made of either LDPE, HDPE, or a mixture of LDPE and HDPE (8–10).

The plastic film surface is almost always treated with one of several alternative surface treatment processes before printing to increase the surface energy of the plastic surface and enhance ink wettability. These surface treatments include flame treatment, corona discharge, plasma treatment, and ultraviolet treatment. Treatment with corona discharge

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Abbreviations: AEO_x, ethoxylated alcohol; AMEO_{x+y}, ethoxylated amine; CMC, critical micelle concentration; DE*, total color difference; DSC, differential scanning calorimetry; EO, ethylene oxide; HDPE, high-density polyethylene; HLB, hydrophilic–lipophilic balance; LDPE, low-density polyethylene; NPEO₁₀, nonylphenol polyethoxylate; TEM, transmission electron microscope.

is common and involves application of high voltage (~10,000 V) to an electrode that is positioned a short distance from the plastic film. The air between the electrode and the plastic is ionized, and this corona of ionized air oxidizes the surface (11). Water-based or oil-based (solvent-based) ink is used for printing the plastic film surface with flexographic or gravure printing processes. Water-based inks occupy a large and increasing fraction of plastic film inks since environmental regulations are restricting the percentage of volatile organic compounds from printing press emissions and the use of water-based inks can reduce capital costs (12–15).

Our previous studies indicated that water-based ink printing could be partially removed from plastic film surfaces with water at pH of 11 or 12 under mechanical action, presumably because the ink binder (acidic acrylate) has a higher solubility at high pH (6). Aqueous solutions of cationic, amphoteric, or nonionic surfactant were identified to be effective deinking agents at pH levels less than 11. Cationic surfactant (i.e., hexadecyl trimethyl ammonium bromide and hexadecyl pyridinium chloride) is as effective at deinking, showing high efficiency at surfactant concentrations both above and below the critical micelle concentration (CMC) over a pH range of 5 to 12. Amphoteric surfactant (i.e., dimethyl dodecylamine oxide) was found to be effective above the CMC over a wide pH range. Nonionic

surfactant (i.e., nonylphenol polyethoxylate) is effective at deinking above the CMC at pH 10 (6).

In this paper, the use of nonionic surfactants of varying structure for plastic film deinking is investigated. Nonionic surfactants are often less expensive and more biodegradable than cationic and amphoteric surfactants. The effects of important parameters are quantified by simple soak and shake experiments. Then two larger-scale apparatuses (a pulper and a flotation cell) are used to simulate pilot-scale deinking.

EXPERIMENTAL PROCEDURES

Materials. Nonylphenol polyethoxylate (NPEO₁₀, trade name Igepal CO-660) was obtained from Rhodia (Cranbury, NJ). Ethoxylated alcohols (AEO_X, trade name Witconol SN-50, SN-70, SN-90, SN-120, 610-30, and TD-100) and ethoxylated amines, which are primary tallow amines (AMEO_{X+Y}, trade name Varonic T-202, T-205, T-210, and T-215), were from the Witco Corporation (Greenwich, CT). Some characteristics of these surfactants are given in Table 1. All surfactants were used as received. The water was double deionized.

The plastic film was composed of 70% LDPE and 30% HDPE. It was reversibly printed with a pattern composed of multicolor water-based ink (black, white, purple, pink, and orange) by flexographic printing after surface treatment with corona discharge. The printed pattern was then coated with

TABLE 1
Properties of Surfactants Used for Deinking of Plastic Film

Acronym (trade name)	Chemical structure	Molecular weight (g/g-mol)	CMC (mM)	HLB	Cloud point (°C) (1 wt%)
Ethoxylated nonylphenol NP(EO) ₁₀ (Igepal CO660)	C ₉ H ₁₉ -C ₆ H ₄ -O(OC ₂ H ₄) ₁₀ H	660	4.9 × 10 ^{-2a}	13.2	60–65 ^b
Ethoxylated alcohol (Witconol): AEO _X = R _n -O(EO) _X H	R _n = C _n H _{2n+1} ; EO = OC ₂ H ₄				
AEO ₃ (SN-50)	R ₁₀₍₁₀₋₁₂₎ -O(EO) ₃ H	296	—	8.9	—
AEO ₅ (SN-70)	R ₁₀₍₁₀₋₁₂₎ -O(EO) ₅ H	387	0.015–0.05% ^c	12	36.6–38.9 ^c
AEO ₇ (SN-90)	R ₁₀₍₁₀₋₁₂₎ -O(EO) ₇ H	446	—	12.9	58.3–60 ^c
AEO ₁₀ (SN-120)	R ₁₀₍₁₀₋₁₂₎ -O(EO) ₁₀ H	576	—	14.3	85–91.7 ^c
A ₆ EO ₃ (610-30)	R ₆₍₆₋₁₀₎ -O(EO) ₃ H	277	—	10	—
A ₁₃ EO ₁₀ (TD-100)	R ₁₃ -O(EO) ₁₀ H	634	—	13.7	66.7–67.8 ^c
Ethoxylated amine (Varonic): tallow primary amine ethoxylate AMEO _{X+Y} = R _n -N-(EO) _X H (EO) _Y H					
AMEO ₂ (T-202)	R _{18(18:1)} -N-(EO)H (EO)H	355–360	—	5.1	—
AMEO ₅ (T-205)	R _{18(18:1)} -N-(EO) ₃ H (EO) ₂ H	470–500	—	9.2	—
AMEO ₁₀ (T-210)	R _{18(18:1)} -N-(EO) ₅ H (EO) ₅ H	680–720	—	12.6	—
AMEO ₁₅ (T-215)	R _{18(18:1)} -N-(EO) ₈ H (EO) ₇ H	900–950	—	14.4	—

^aFrom Reference 35.

^bFrom Reference 27.

^cFrom Witco Detailed Product Report (Witco Corporation). CMC, critical micelle concentration; HLB, hydrophilic–lipophilic balance.

white ink to give an opaque appearance and to protect the ink layers (6). This film was commercial plastic packaging supplied by the Kimberly-Clark Corporation (Neenah, WI). The clear and the printed parts of the plastic film were segregated for this study, and the clear parts of plastic film were used as a standard for the comparison with deinked plastic film strips.

Methods. All glassware was cleaned with a regular lab detergent and then acid-washed with Nochromix® solution (Godax Laboratories Inc., Takoma Park, MD) for at least an hour. Surfactant solutions were always prepared fresh with double-deionized water for each experiment, and the pH level of the aqueous surfactant solution was adjusted immediately with sodium hydroxide or hydrochloric acid before the deinking experiment was conducted. A representative sample printed with multicolors of water-based ink (white, black, orange, purple, and pink) was cut from the printed area of the plastic packaging to the size of 1 × 1 in. unless otherwise specified and soaked in the surfactant solutions. The effect of process parameters was studied in experiments performed with 20 mL of surfactant solutions including 20 irregularly shaped porcelain beads in 30-mL glass vials. The glass vials were placed in a water bath at a monitored temperature when the effect of soaking prior to agitation was studied. These samples were agitated for varied time periods with a portable reciprocating action shaker (Model R-2; New Brunswick Scientific, Edison, NJ) at the speed of 350 strokes per minute with a 1-in. stroke length. After agitation, plastic film strips were taken out of the washing bath, rinsed with deionized water, and air dried. A shaker was placed in an incubator (Fisher Scientific) when temperatures other than room temperature were used. In addition, several experiments to demonstrate scaleup were performed with 2 L of surfactant solutions by using either a Denver Laboratory Model D-12 Flotation Machine with the attrition scrubber kit (Denver Equipment Company, Denver, NC) or a TMI D-111 disintegrator (Testing Machines Inc., Islandia, NY). The Denver flotation cell (Fig. 1) is commonly used for pilot-scale paper deinking studies to separate ink from paper fibers after pulping. In these flotation deinking experiments, ink particles attach to air bubbles and are collected as froth from the top of the solution. For this work, the Denver flotation cell was converted to a scrubber by removing the air valve and replacing the impeller with a one-piece double-attrition propeller. The agitation speed of this machine can be adjusted between 500 and 3500 rpm. The TMI disintegrator (Fig. 2) is a pulper used for pilot-scale paper deinking to disintegrate waste paper. Besides agitation, the TMI disintegrator (Fig. 2) provides a beating effect due to the attached spiral fins on the internal part of the pulping tank. The agitation speed of the disintegrator is limited to 3000 rpm. After deinking experiments with these machines, the washing bath was drained, and deinked plastic film strips were rinsed several times with water and dried in a vacuum oven overnight.

Each deinked plastic film sample was first visually evaluated and then analyzed using a HunterLab UltraScan™ Spectrocolorimeter system (Hunter Associates Laboratory, Inc., Reston, VA) in regular transmission mode. The HunterLab

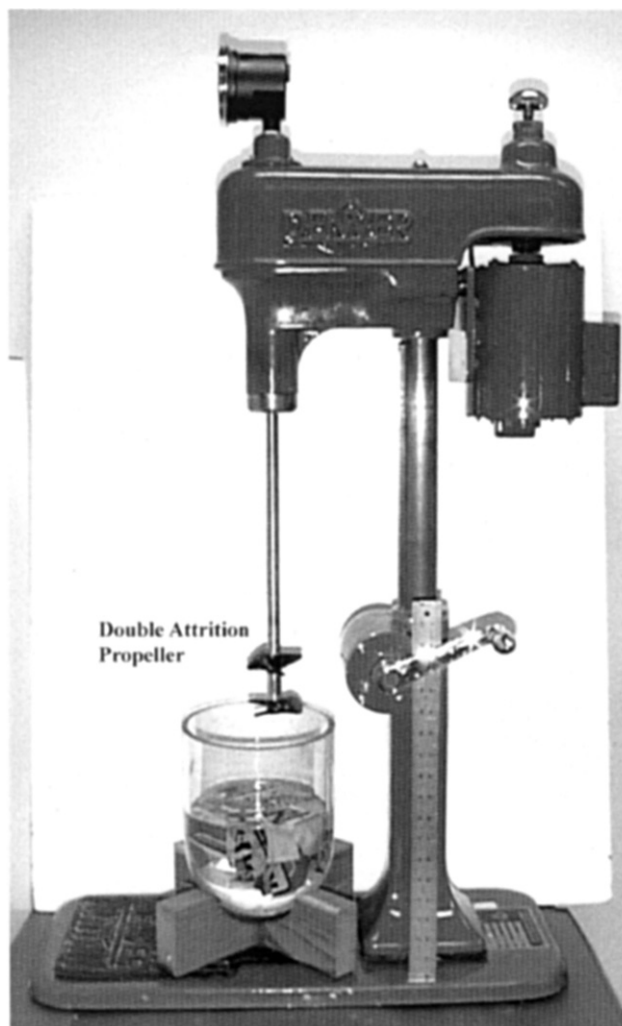


FIG. 1. Denver flotation cell (Denver Equipment Co., Denver, NC).

UltraScan Spectrocolorimeter system consists of an UltraScan optical sensor (SN-7834) with a sensor interface unit and an IBM computer including data evaluation software. In the regular transmittance mode, the intensity of the light passing directly through the plastic film is measured as a function of wavelength. The total color difference (DE^*) between the standard (clear plastic film) and the deinked sample was determined at the chosen standard illuminant (D65, representing daylight with a correlated color temperature of 6500 K) and observer (10° standard observer). An average value from six measurements for each sample is reported. The lower the value of DE^* , the better the deinking. More details about this procedure are given in Reference 6.

Clear plastic film, printed plastic film, and deinked plastic film were separately extruded either as pellets or as a film with a KL-100 Killion Extruder at 232°C. Processing temperatures of 160–240°C for LDPE and 200–280°C for HDPE are recommended (9). Cross-sectional areas of the re-extruded pellets were analyzed with a transmittance electron microscope (TEM; JEOL 2000 FX, Peabody, MA) and energy-dispersive X-ray spectrometer attached to the TEM. Samples for the TEM were prepared with 600 Ultramicrotomy

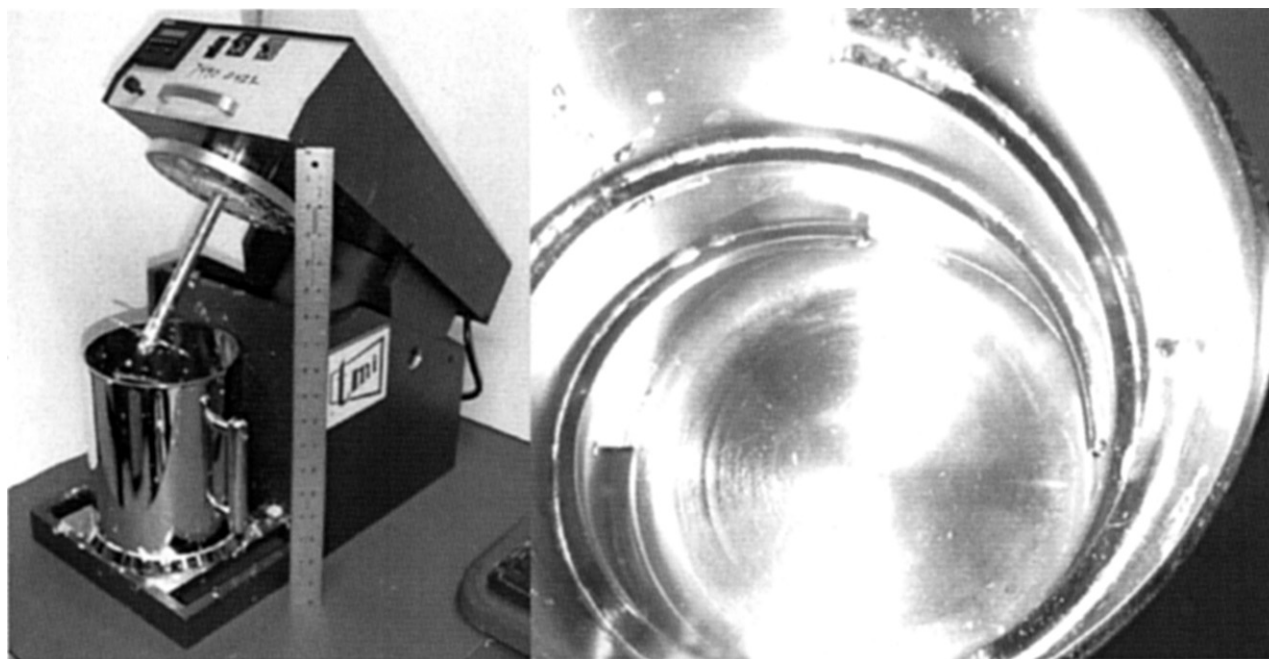


FIG. 2. TMI disintegrator (Testing Machines Inc., Islandia, NY).

attached with FS-100 Cryosectioning (Microstar Technologies, Huntsville, TX) at -120°C and picked up with 200-mesh nickel or copper folding grids.

For stress-strain property measurements, re-extruded film strips were first pressed with a Carver M Press (Carver Inc., Wabash, IN) including a Carver 2101 heating plate at approximately 150°C in order to achieve uniform thickness and then cut with an ASTM standard D-1708 die (16). The samples were stretched with an Instron TT-C-L tensile tester (Instron Co., Canton, MA), which had a computerized data acquisition system, at a rate of 0.51 cm/min at room temperature. Measurements were repeated three times for re-extruded clear, printed, or deinked film. The differential scanning calorimetry (DSC) data for pellets produced from clear, printed, and deinked plastic films were obtained using a liquid nitrogen-cooled Mettler-Toledo DSC820 (Columbus, OH). Samples were placed in 40- μL aluminum pans. Thermograms were recorded from 20 to -150°C and from -150°C to 150°C at a heating rate of -10 and $5^{\circ}\text{C}/\text{min}$, respectively. From the DSC curve, the melting temperature (T_m) and the heat of fusion, which is the melting peak area, were obtained. The crystallinity of the polymer was calculated by dividing the measured heat of fusion by the heat of fusion of 100% crystalline polyethylene (290 J/g). Polymer produced from the printed part of the plastic film had 8.6 wt% ink; therefore, the crystallinity of this sample was corrected by dividing the calculated crystallinity by the mass fraction of polymer (0.914).

The densities of the re-extruded plastic films were measured according to ASTM standard D-1505 (17) by use of the DC-4 Techne Density Gradient apparatus (Techne Incorporated, Princeton, NJ). This test method was based on observing the level of plastic pellets in a density gradient liq-

uid column and then comparing it with standard glass floats of known density. An isopropanol/water (50:50 vol%) mixture was used as the liquid in the density gradient column.

The procedure of ISO 1065, method A—Determination of the cloud point in aqueous solution (18), was followed for the measurement of the cloud points of AMEO₂, AMEO₅, AMEO₁₀, and AMEO₁₅. This method does not monitor the effect of pH on the cloud point. For this study, the pH level of the surfactant solution was adjusted between 4 and 11 using sodium hydroxide or hydrochloric acid. A surfactant solution (15 mL of 1 wt%) in a test tube was placed in a water bath filled with ethylene glycol. The temperature was increased to 100°C and then decreased until the cloudy appearance of the solution disappeared. This temperature was recorded as the cloud point of the surfactant.

To determine whether AMEO₅ protonates, titration (60 mL of 5 mM, prepared based on the maximum value of its molecular weight) was performed with 0.25 N Fisher-brand sodium hydroxide and hydrochloric acid. Eppendorf micropipettes, 100 and 1000 μL , were used to add a defined amount of acid or base into the surfactant solution, and the pH of the solution was monitored with an Accumet model 50 pH meter (Fisher Scientific, Fair Lawn, NJ). Surfactant solution was first titrated with hydrochloric acid to pH 2 or with sodium hydroxide to pH 12. The solution titrated to pH 12 was back-titrated with hydrochloric acid to pH 2. A similar titration was also performed using water for comparison of titration curves.

RESULTS AND DISCUSSION

Quantification of deinking process parameters. Deinking involves the removal of, first, white ink coating and then colored ink

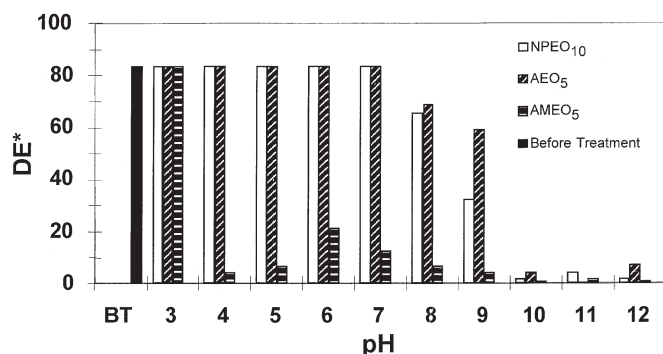


FIG. 3. The total color difference (DE^*) between the deinked plastic film using either $NPEO_{10}$, AEO_5 , or $AMEO_5$ and clear plastic film as a function of pH. NPEO, nonylphenol polyethoxylate; AEO, ethoxylated alcohol; AMEO, ethoxylated amine.

layers from plastic film. Surfactants $NPEO_{10}$, AEO_X , and $AMEO_{X+Y}$ (some characteristics of these are given in Table 1) were chosen as model nonionic surfactants for this study. The solution of $NPEO_{10}$ was identified to be effective at deinking above the CMC at room temperature and pH 10 after 4 h of agitation (6). There is substantial controversy over the toxicity and environmental impact of the $NPEO_X$ class of nonionic surfactants (19–22).

Ethoxylated alcohols are nonionic surfactants that are extensively used in the paper-deinking processes (23). They are relatively inexpensive and exhibit excellent wetting and detergency properties, especially for oily soil removal (24). An ethoxylated alcohol, AEO_5 (whose properties are shown in Table 1), was tested in deinking of plastic packaging at a concentration of 25 mM, at room temperature, at pH from 3 to 12. The resulting DE^* values are presented in Figure 3, and a decrease in DE^* values indicates a decrease in printed ink residue on plastic film, i.e., an increase in the amount of deinking. As seen in Figure 3, the deinking level achieved with AEO_5 is comparable to that achieved with $NPEO_{10}$. No deinking was achieved with AEO_5 at acidic and neutral pH levels, but the deinking process started at basic pH level and increased with increasing pH. Therefore, ethoxylated alcohols can be effective deinking agents at a process pH of 10 or greater.

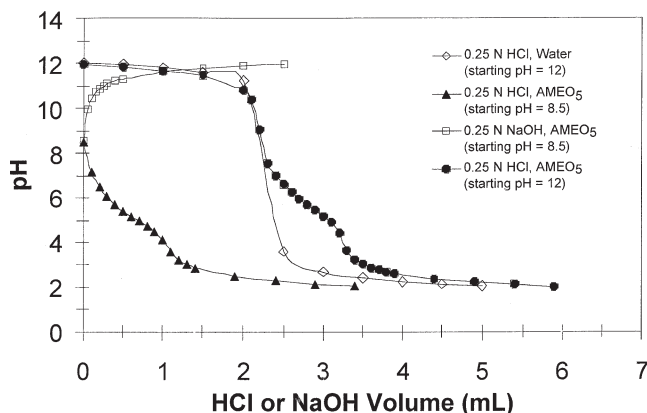


FIG. 4. Titration of $AMEO_5$. For abbreviation see Figure 3.

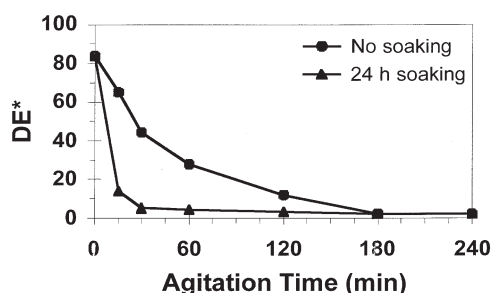


FIG. 5. DE^* between the deinked plastic film using 25 mM $NPEO_{10}$ [room temperature (room T) and pH = 10] and clear plastic film as a function of agitation time. For abbreviations see Figure 3.

Ethoxylated amines are classified as nonionic surfactants, but they behave as cationic surfactants in acid solution and at low ethylene oxide levels owing to the presence of a nitrogen atom (24,25). For example, the titration of $AMEO_5$ with hydrochloric acid, shown in Figure 4, demonstrates that $AMEO_5$ protonates under acidic conditions. Ethoxylated amines exhibit good emulsification and wetting properties at both acidic and alkaline pH levels, and are used as dye leveling agents in the textile industry and as dispersing agents for pigments in oil-based paints (23,24). The $AMEO_5$ (properties shown in Table 1) was tested for the deinking of plastic film. Experiments were performed with 5 mM $AMEO_5$ at room temperature by varying the pH levels from 3 to 12. Unlike $NPEO_{10}$ and AEO_5 , $AMEO_5$ (Fig. 3) accomplishes deinking at acidic, neutral, and basic pH levels, and residual ink on plastic film decreases with an increase in acidity or alkalinity. This result is comparable to that obtained with cationic surfactant (6). Hence, ethoxylated amines also can be used effectively for deinking of plastic film over a wide pH range.

The effect of agitation period on deinking was investigated with or without soaking the plastic strips in surfactant solution prior to agitation. Experiments were performed with 25 mM $NPEO_{10}$ at room temperature at a pH of 10. Agitation time was varied from 0 to 240 min. The resulting DE^* values are presented in Figure 5. The residual ink on film decreases as the agitation time increases from 0 to 180 min and stabilizes at 180 min when the plastic film strips are agitated without soaking them in the surfactant solution

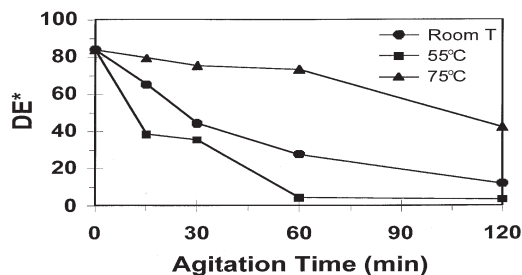


FIG. 6. DE^* between the deinked plastic film using 25 mM $NPEO_{10}$ (pH = 10) and clear plastic film as a function of agitation time at varied temperatures (plastic film was not soaked in surfactant solution prior to agitation). For abbreviations see Figures 3 and 5.

prior to the agitation step. However, pre-soaking the plastic film in a surfactant solution for 24 h significantly decreases the residual ink on film for a given agitation time, but the value at which the DE^* stabilizes at long times is identical in the two cases.

The effect of temperature on the deinking process was also investigated with NPEO₁₀ and the resulting DE^* values are presented in Figures 6 (samples were agitated without pre-soaking) and 7 (plastic film was pre-soaked in surfactant solution for 24 h). Experiments were conducted with 25 mM NPEO₁₀ at a pH of 10. As seen in Figure 6, the residual ink on film decreases as the temperature is increased from room temperature (~25°C) to 60°C, but less ink is removed when the temperature is increased to 75°C. Nonionic surfactants become turbid when they are heated to the cloud point, a temperature at which the solution separates into two phases, the micelle-rich phase and the micelle-poor phase (26). As seen in Table 1, NPEO₁₀ has a cloud point of 60–65°C (27). If it were not for a phenomenon such as a phase change, deinking would be expected to improve monotonically with increasing temperature. The increase in DE^* after the minima as a function of temperature shown in Figure 7 seems likely to be related to the cloud point being exceeded. In laundering, optimal removal of nonpolar oil phases from fabrics occurs if the washing bath temperature is 15–25°C above the surfactant cloud point. However, the optimal removal of polar oil phases or blends of nonpolar and polar oil phases from fabric occurs at wash temperatures 0–30°C below the surfactant cloud point (28). Ink appeared to behave like polar oil in the paper deinking process (28,29). Although the optimal deinking temperature was not determined with accuracy, an optimal temperature in the vicinity of the cloud point is consistent with these guidelines for fabric detergency behavior.

The HLB value is used to characterize surfactants and is calculated from the structure of the molecule or obtained from experimental emulsification data (30,31). The higher the HLB value, the more water-soluble it tends to be and the higher the cloud point tends to be.

The effect of the HLB of ethoxylated alcohols and ethoxylated amines on deinking at room temperature was studied here. The HLB of the ethoxylated alcohols was varied by changing either the number of ethylene oxide (EO)

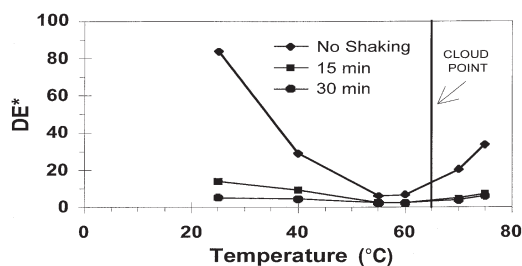


FIG. 7. DE^* between the deinked plastic film using 25 mM NPEO₁₀ (pH = 10) and clear plastic film as a function of temperature at varied agitation times (plastic film was soaked in surfactant solution for 24 h at process temperature prior to agitation). For abbreviations see Figure 3.

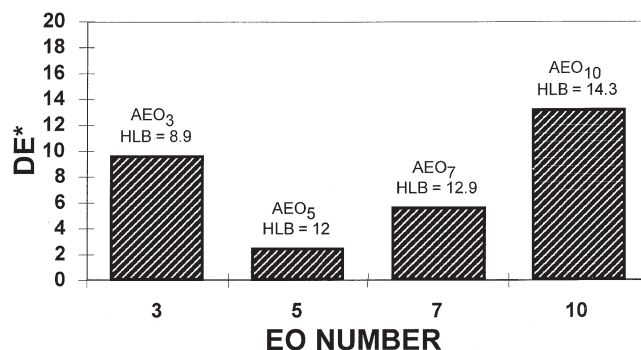


FIG. 8. DE^* between the deinked plastic film using AEO_x and clear plastic film as a function of ethylene oxide (EO) numbers (room T and pH = 10). HLB, hydrophilic-lipophilic balance; for other abbreviations see Figures 3 and 5.

groups (EO number changes from 3 to 10 while the carbon chain length is kept constant at about 10) or its carbon chain length (6, 10, 13, and nonylphenol while keeping EO number at either 3 or 10). The HLB of the ethoxylated amines was varied by changing only the number of EO groups from 2 to 15. Increasing the number of EO groups increases the HLB number, so the surfactant becomes more water soluble (Table 1). Adding EO groups to the surfactant also elevates the cloud point of the surfactant. On the other hand, increasing the carbon chain number decreases the HLB number and cloud point of the surfactants (Table 1).

Plastic packaging strips were soaked for 24 h in 20 mL of 25 mM AEO₃, AEO₅, AEO₇, AEO₁₀, A₆EO₃, A₁₃EO₁₀, and NPEO₁₀ (each solution had only one plastic strip of 1 × 1 in. dimensions) at pH 10, then 20 porcelain abrasives were added to the solutions and agitated for 15 min. As one can see in Figure 8, decreasing the EO number increased the deinking capacity of the surfactant except for AEO₃. The solution of AEO₃ was the only one in which a milky phase appeared, indicating that the surfactant cloud point is below room temperature. Extrapolation of the cloud point data (Witco Detailed Product Report, 1 wt%, pH = 6.5–7) in Table 1 results in an estimated cloud point of AEO₃ of 15°C. The higher level of deinking was achieved with AEO₅ (Fig. 8), which is consistent with an optimal temperature being

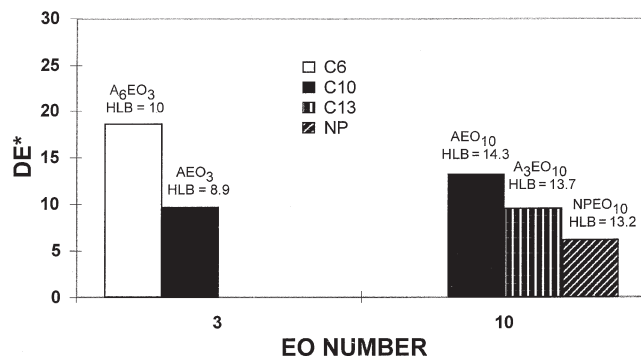


FIG. 9. DE^* between the deinked plastic film using nonionic surfactants with varied carbon chain length and clear plastic film as a function of EO numbers (room T and pH = 10). C represents carbon molecule; NP is nonylphenol; for other abbreviations see Figures 3, 5, and 8.

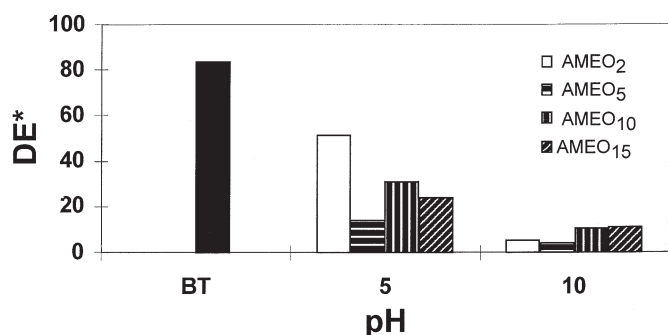


FIG. 10. DE* between the deinked plastic film using AMEO_x and clear plastic film as a function of EO numbers (room T and pH = 5 and 10). For abbreviations see Figures 3, 5, and 8.

around the cloud point. As seen in Figure 9, increasing the carbon chain length also increased the deinking capacity of the surfactant.

The HLB of the ethoxylated amines was varied by changing the EO number. The ethoxylated amines studied here were reported to have no cloud points between 0 and 100°C (32). However, the cloud point for AMEO₅ was determined as 40.5°C at pH 8, 28°C at pH 9, 26.5°C at pH 10, and 25.5°C at pH 11. The cloud point for AMEO₁₀ at pH 10 and 11 was 98°C. At temperatures between 1 and 100°C, AMEO₂ was cloudy at pH 5 to 11. However, AMEO₂ at pH 4, AMEO₁₀ at pH 4 to 9, and AMEO₁₅ at all pH levels formed clear solutions. Deinking experiments were performed at room temperature and at pH values of 5 and 10 by first soaking plastic packaging samples in 5 mM surfactant solutions for 24 h and then by shaking the samples for 15 min. The resulting DE* values are presented in Figure 10. As one can see in Figure 10, the deinking capacity of the surfactant increases with decreasing EO number (similar to Fig. 8) except for AMEO₂. Better deinking was achieved with AMEO₅ at the highest pH studied.

Since AMEO₅ is effective at deinking of water-based ink at acidic, neutral, or alkaline pH levels (see Fig. 3), the effect of agitation period at deinking with or without soaking the plastic strips in surfactant solution was investigated. Experiments were performed with a surfactant concentration of 5 mM

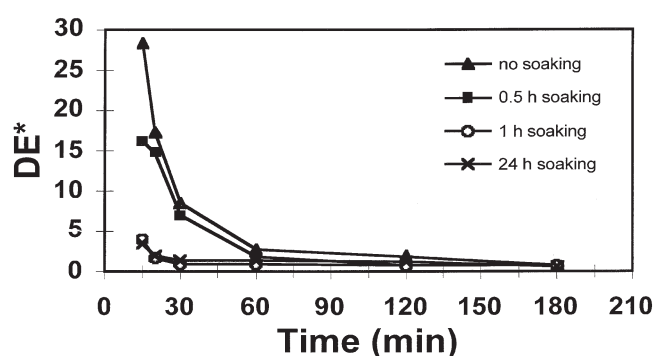


FIG. 12. DE* between clear plastic and water-based ink printed plastic film treated with 5 mM AMEO₅ at pH 10 as a function of agitation time at varied soaking periods. For abbreviations see Figure 3.

AMEO₅ at room temperature at pH of 5, 7, and 10. Agitation time was varied from 0 to 240 min. As seen in Figure 11, the residual ink on film decreased (deinking level increased) as agitation time increased from 0 to 60 min and stabilized at 60 min. AMEO₅ deinks more effectively if the surfactant solution is acidic (pH 5) or basic (pH 10) than if it is neutral.

The effect of soaking on deinking prior to mechanical agitation was also investigated with 5 mM AMEO₅ at pH 10. Soaking plastic samples in surfactant solution (Fig. 12) increased the deinking level and decreased the needed agitation time for the optimal level of deinking. As seen in Figure 12, increasing the soaking period from 30 min to 1 h decreased greatly the residual ink on film (increased the deinking level); however, further increase of the soaking period from 1 h to 24 h did not cause any significant change on the deinking level. On the basis of the information in Figures 5 and 12, long soaks (>1 h) prior to agitation do not benefit deinking substantially, especially with extended agitation times.

The effect of plastic film consistency (plastic/solution ratio) on deinking was studied by varying it from 0.1 to 5% using 20 mL of 5 mM AMEO₅ at pH 10. Plastic film strips were soaked in the surfactant solution for 1 h prior to an agitation period of 20 min. As seen in Figure 13, an increase in consistency increased the residual ink on film. The amount of residual ink on film was also varied from one plastic strip to another in the same batch when the consis-

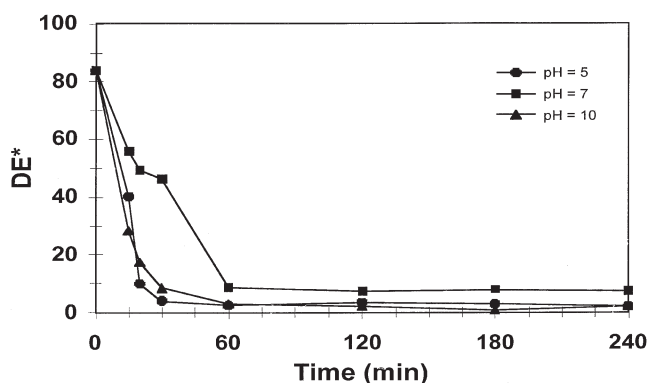


FIG. 11. DE* between clear plastic and water-based ink printed plastic film treated with 5 mM AMEO₅ as a function of agitation time at varied pH values. For abbreviations see Figure 3.

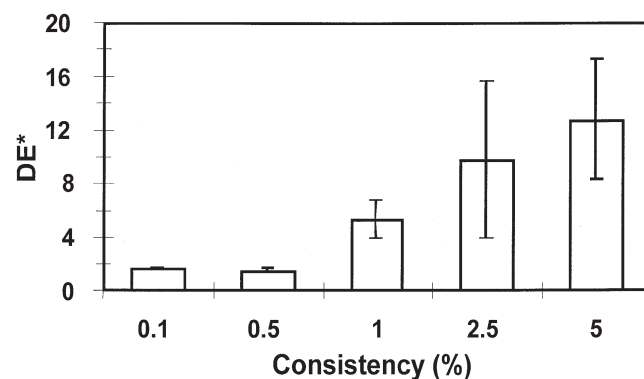


FIG. 13. DE* between clear plastic and water-based ink printed plastic film treated with 5 mM AMEO₅ at pH 10 as a function of consistency (plastic/solution ratio). Error bars represent the 90% confidence interval. For abbreviations see Figure 3.

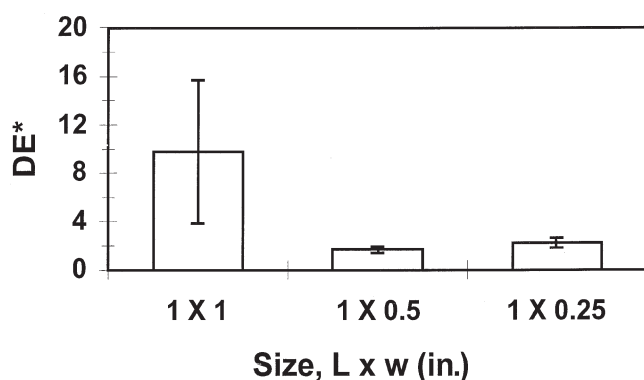


FIG. 14. DE* between clear plastic and water-based ink printed plastic film treated with 5 mM AMEO₅ at pH 10 with 2.5% consistency (plastic/solution ratio) as a function of plastic film size. Error bars represent the 90% confidence interval. For abbreviations see Figure 3.

tency was 1, 2.5, or 5%. However, the size of the plastic film strips affected the level of deinking. Three experiments were performed with 2.5% consistency. Plastic film for each experiment was cut to 1 × 1, 1 × 0.5, or 1 × 0.25 in. As seen in Figure 14, a decrease in the width of the plastic film strip from 1 to 0.5 in. decreased the residual ink on film and increased the uniformity of the deinking among different samples. However, a decrease in width from 0.5 to 0.25 in. did not significantly change the deinking level. Error bars in Figures 13 and 14 represent the 90% confidence interval (for each batch, the DE* values of all the deinked plastic film strips were determined. The mean and lower/upper limits of the error bar for the DE* values were calculated with the method stated in Ref. 33). The size of plastic film strips in the deinking process may be important in the recovery of plastic film pieces after the deinking operation.

Visual evaluation of the deinked plastic film samples indicated that black ink is more difficult to remove than other colors studied here.

Scaleup of deinking. Deinking experiments were scaled up from 20 mL to 2 L by using either a Denver flotation cell or a TMI disintegrator. The surfactant concentration was kept at 5 mM and the process pH was 10 with surfactants AEO₅ and AMEO₅. The printed part of the plastic film at 1% con-

sistency (20 g plastic film in 2000 mL solution) was first cut to the size of 1 × 1 in., soaked in the surfactant solution for 1 h, and then agitated in the Denver flotation cell or the TMI disintegrator for 20 min. The agitation speed of the Denver flotation cell was varied from 1000 to 3000 rpm. The experiment with AEO₅ was performed at only 1000 rpm with the Denver flotation cell because AEO₅ created a large amount of foam during agitation. After deinking, the plastic film pieces were collected, rinsed with water, and dried in a vacuum oven overnight. From each batch, 30 plastic film pieces (around 3 wt% of the total plastic film) were arbitrarily picked and analyzed with the HunterLab UltraScan Spectrocolorimeter. Figure 15 shows the average DE* between clear plastic film and deinked plastic film (with a 90% confidence interval) as a function of agitation speed. Increasing the agitation speed from 1000 to 3000 rpm with the Denver flotation cell decreased the relative residual ink amount on film when AMEO₅ was used as a surfactant. The Denver flotation cell and the TMI disintegrator at 3000 rpm resulted in almost the same level of deinking, but, owing to the beating effect, using the TMI disintegrator provided a more uniform deinking among plastic film pieces than using the Denver flotation cell. The plastic film deinks better with AEO₅ than with AMEO₅ at 1000 rpm when the Denver flotation cell is used (Fig. 15). However, foam that formed with the use of AEO₅ created several problems during the agitation and rinsing stages. Addition of defoamer to the AEO₅ solution might eliminate these problems. Furthermore, experimental observation indicated that AMEO₅ acted as a flocculant since the majority of the particles from the washing bath settled down. The agitation speed affected the settling period of the ink particles. For the washing bath that was collected from the plastic film deinked at 3000 rpm, complete ink particle settling took 24 h. However, settling took 3 and 7 d for washing baths collected from the plastic film deinked at 2000 and 1000 rpm, respectively. On the other hand, AEO₅ acted as a dispersant because the majority of the ink particles stayed suspended in the washing bath after the deinking.

Hence, these experiments, performed with either a Denver flotation cell or a TMI disintegrator, show that deinking of plastic film using these surfactants is technically feasible.

Physical properties of re-extruded plastic films. Shredded clear or printed parts of the plastic film packaging and plastic film strips deinked with either AMEO₅ at 3000 rpm with the TMI disintegrator or AEO₅ at 1000 rpm with the Denver flotation cell were re-extruded either as pellets or as a film. Clear plastic film had a beige appearance (the natural color of a PE resin is clear or milky white) upon re-extrusion into pellets, whereas re-extruded printed film was reddish brown. Clear plastic film is film that has not been printed on but that has been treated with a corona discharge, which leaves reactive functional groups (i.e., hydroxyl groups, ethers, ketones, aldehydes, carboxylic acid groups, and carboxylic ester groups) on the film surface. These functional groups are probably the cause of the degradation of optical properties reflected in the beige color upon re-extrusion of

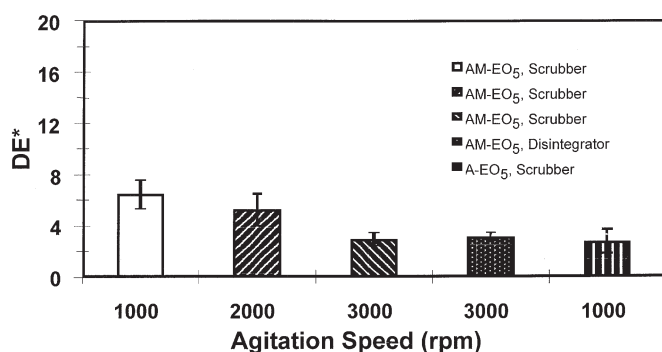


FIG. 15. DE* between clear plastic and water-based ink printed plastic film treated with 5 mM AMEO₅ and AEO₅ and at pH 10 at large scale (2 L) as a function of agitation speed. Error bars represent the 90% confidence interval. For abbreviations see Figure 3.

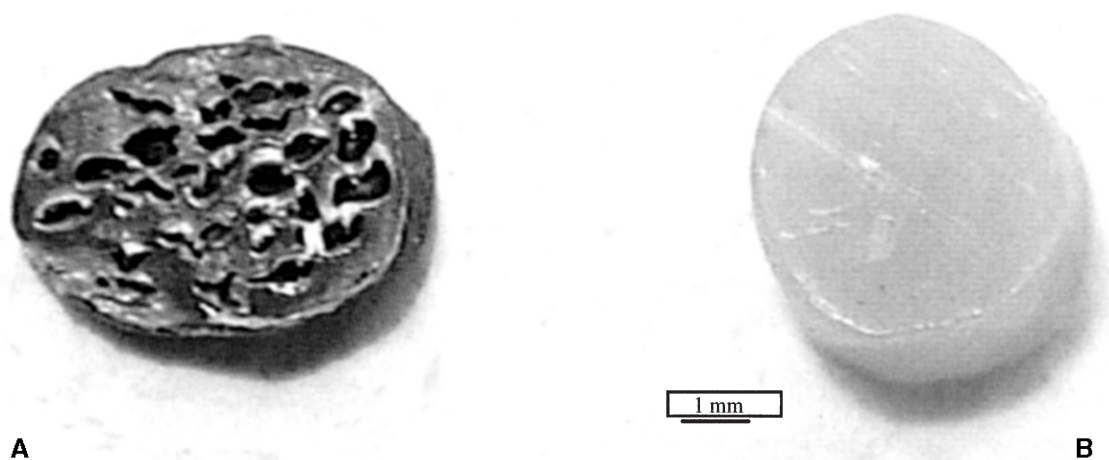


FIG. 16. Cross-sectional area of repelletized clear (B) and printed (A) plastic film.

clear film. As seen in Figure 16, crack points and holes were observed on the cross-sectional area of the pellets produced from printed film. TEM micrographs and X-ray charts of re-extruded clear, printed, and deinked films with AMEO₅ and AEO₅ are shown in Figures 17 to 20, respectively. The difference between these micrographs is the presence of ink aggregates of approximately 2–5 μm in size seen in Figure 18. These ink aggregates disappeared when the plastic film

was treated with either AMEO₅ or AEO₅ (Figs. 19, 20). Unremoved ink particles stayed in the plastic structure as either single particles or as much smaller aggregates. As seen in Figure 18, titanium, which is commonly found in white ink in the form of TiO₂, was detected in ink aggregates with X-ray spectrometric measurements. The counts of titanium decreased 89% when plastic film was treated with AMEO₅ and 94% when plastic film was treated with AEO₅ (Figs. 19,

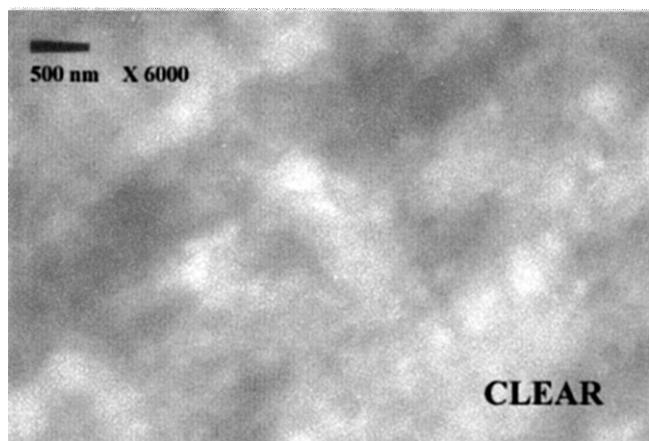


FIG. 17. Transmission electron microscope (TEM) micrograph and X-ray chart of re-extruded clear plastic film. C, carbon; O, oxygen; Ni, nickel.

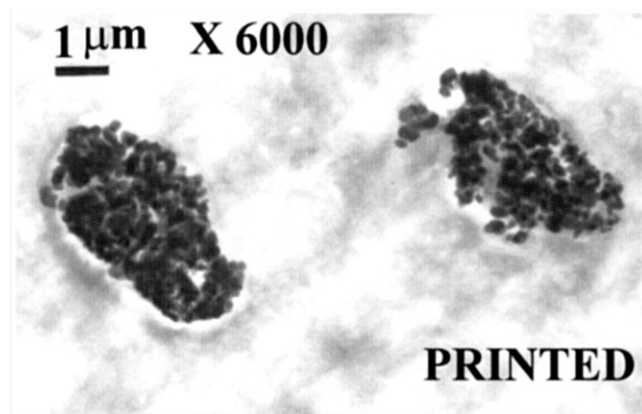


FIG. 18. TEM micrograph and X-ray chart of re-extruded printed plastic film. Ti, titanium; for other abbreviations see Figure 17.

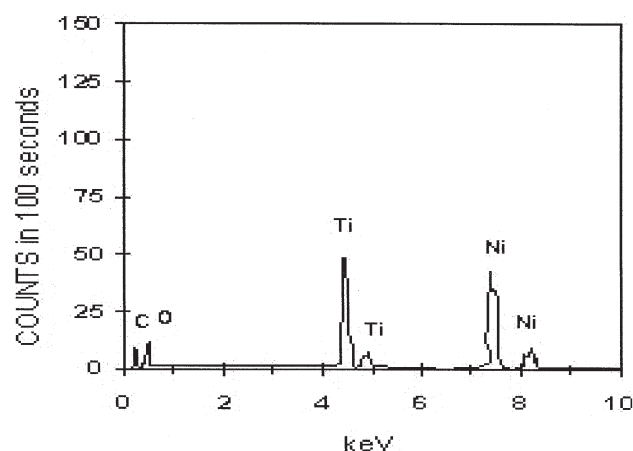
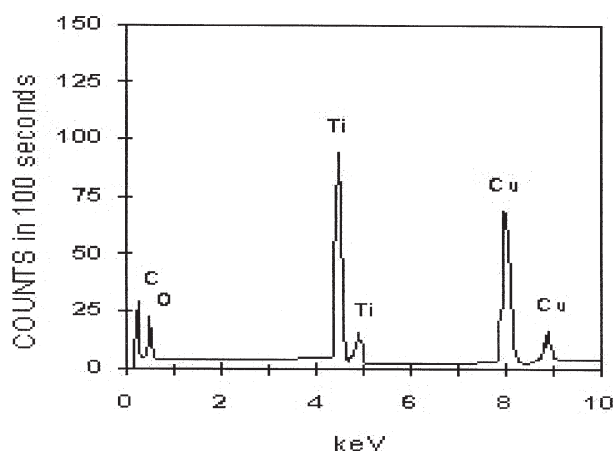
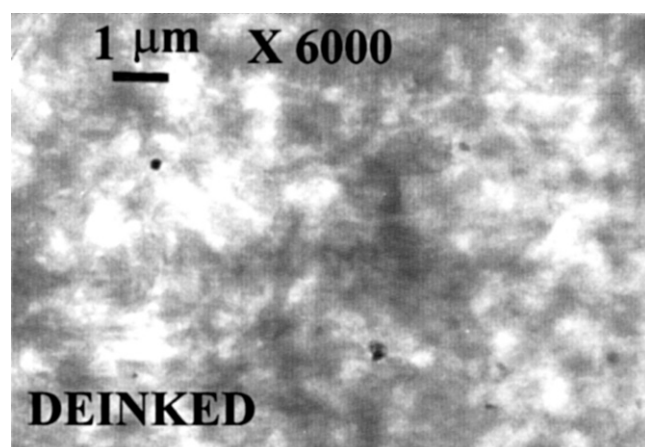
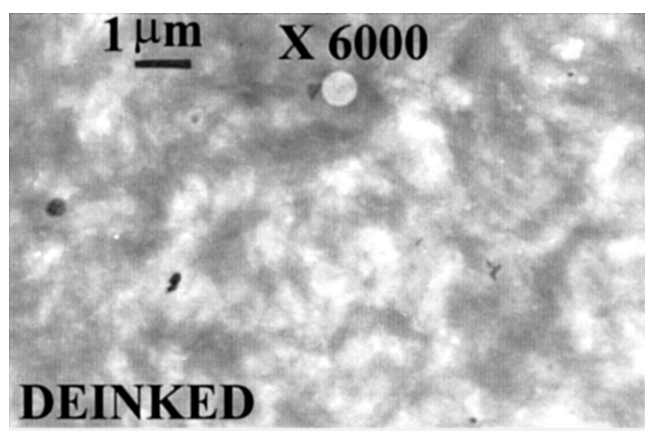


FIG. 19. TEM micrograph and X-ray chart of re-extruded deinked plastic film using AMEO₅ with disintegrator at 3000 rpm. For abbreviations see Figures 3, 17, and 18.

FIG. 20. TEM micrograph and X-ray chart of re-extruded deinked plastic film using AEO₅ with flotation cell at 1000 rpm. For abbreviations see Figures 3, 17, and 18.

TABLE 2
Physical Properties of Re-extruded Plastic Films^a

Properties	Clear plastic	Printed	Deinked with AMEO ₅	Deinked with AEO ₅
Modulus (MPa)	486.53 ± 52.15	289.99 ± 25.27	442.45 ± 58.01	433.71 ± 79.71
Yield stress (MPa)	16.78 ± 0.56	15.79 ± 0.51	15.93 ± 0.90	16.45 ± 0.50
Ultimate strength (MPa)	16.84 ± 0.44	12.93 ± 1.08	16.25 ± 0.85	16.99 ± 1.16
Elongation at break (%)	634 ± 115	192 ± 12	551 ± 47	596 ± 45
Crystallinity (%)	53.2 ± 19.4	57.3 ± 10.7	50.2 ± 0.8	48.2 ± 0.6
T _m (°C)	117.3 ± 4.8	117.7 ± 1.2	118.9 ± 0.5	119.5 ± 1.5
Density (g/cm ³)	0.9378	0.9564	0.9358	0.9343

^aT_m, melting temperature; for other abbreviations see Table 1.

20). In these charts, detected nickel and copper peaks represent the sample folding grids.

Ink aggregates in the plastic structure are expected to reduce the strength of the plastic. Particle agglomeration is a weak point in the material and causes the material to break easily when stress is applied to it (34). Therefore, the stress-strain properties of re-extruded samples from clear, printed, and deinked plastics were determined. Stress-induced crystallinity was observed during stretching of all samples. Measurements were repeated three times for each sample, and the stress-strain properties with a 90% confi-

dence interval are presented in Table 2. Ink makes the plastic less stiff (modulus is decreased) and weaker (ultimate strength and elongation at break are decreased). Therefore, plastic heavily contaminated with printing is not a suitable raw material for clear plastic film production due to poorer physical properties and color from residual ink. However, deinking printed plastic film with either AMEO₅ or AEO₅ changes the color; reduces the density; and raises the modulus, the ultimate strength, and the elongation at break to a level basically equivalent to that of clear plastic film (Table 2). On the other hand, ink in repelletized poly-

mer or the treatment with surfactants does not change the fractional crystallinity or melting temperature of the polymer (see Table 2).

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